

Solutions to Problem Set 4

1. (*) Benford's law and σ -algebras

[4 Points]

Empirical evidence suggests that the first digits in lists of large data sets of real numbers appearing naturally (such as natural constants, lengths of rivers, stock prices, ...) are *not* uniformly distributed on $\{1, \dots, 9\}$. Instead, they follow the so-called Benford distribution, which is a discrete distribution on $\{1, \dots, 9\}$ with

$$p(k) = \log_{10} \left(1 + \frac{1}{k} \right), \quad k \in \{1, \dots, 9\}$$

(Newcomb 1881, Benford 1938). We want to make this mathematically plausible, and aim at constructing a probability measure $\mathbf{P}$ on $(0, \infty)$ equipped with an appropriate σ -algebra $\mathcal{M}$ with

$$(*) \quad \mathbf{P}[\{\omega \in (0, \infty) : \text{the decimal representation of } \omega \text{ starts with digit } k\}] = \log_{10} \left(1 + \frac{1}{k} \right).$$

It is clear that $\mathbf{P}$ has to be scale invariant for all $\alpha > 0$, i.e. $\mathbf{P}[\alpha B] = \mathbf{P}[B]$ for all $\alpha > 0$ and $B \in \mathcal{M}$.¹ One can show that for $\mathcal{B}((0, \infty)) = \{B \cap (0, \infty) : B \in \mathcal{B}(\mathbb{R})\}$, no such probability measure exists. However, one can consider a smaller σ -algebra

$$\mathcal{M} = \sigma(\mathcal{E}), \text{ where } \mathcal{E} = \{M_t : 1 < t < 10\}, \text{ with sets } M_t = \bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, t).$$

One can then show that the expression $\mathbf{P}[M_t] := \log_{10}(t)$ can be extended to a probability measure $\mathbf{P}$ on $((0, \infty), \mathcal{M})$ (you do not need to show this).

(a) Show that $\alpha M_t \in \mathcal{M}$ for all $\alpha > 0$ and all $M_t \in \mathcal{E}$.

(b) Show that $\mathbf{P}[\alpha M_t] = \mathbf{P}[M_t]$.

(c) Show that $(*)$ holds.

(d) Show that the properties of (a) and (b) hold for all $M \in \mathcal{M}$.

Hint: Show that $\mathcal{M}' = \{M \in \mathcal{M} : \alpha M \in \mathcal{M}\}$ is a σ -algebra. You can use without proof that $\mathcal{M}'' = \{M \in \mathcal{M} : \mathbf{P}[\alpha M] = \mathbf{P}[M]\}$ also a σ -algebra. Why does this imply the result?

(a,b) Suppose without loss of generality that $\alpha \in (1, 10)$.

Case 1 : Let $\alpha t < 10 \implies$

$$[\alpha, \alpha t) = [1, \alpha t) \setminus [1, \alpha) \in \mathcal{M}$$

and

$$\mathbf{P}[\alpha M_t] = \log_{10} \alpha t - \log_{10} \alpha = \log_{10} t$$

Case 2 : Let $\alpha t > 10 \implies$

$$\begin{aligned} [\alpha, \alpha t) &= [\alpha, 10) \cup [10, \alpha t) \\ \implies \bigcup_n [\alpha, \alpha t) \cdot 10^n &= \left(\bigcup_n [\alpha, 10) \cdot 10^n \right) \cup \left(\bigcup_n \left[1, \alpha \frac{t}{10} \right) \cdot 10^n \right) \in \mathcal{M} \\ \implies \mathbf{P}[\alpha M_t] &= 1 - \log_{10} \alpha + \log_{10} \left(\alpha \frac{t}{10} \right) \\ &= 1 - \log_{10} \alpha + \log_{10} (\alpha) + \log_{10} (t) - 1 \\ &= \log_{10} t \end{aligned}$$

¹This is since the quantity of interest is, say, a length, and it should be invariant under changing the unit of measurement.

(c) One has, for $k \in \{1, \dots, 9\}$

$$\begin{aligned} \mathbf{P}[\{\text{first digit of } \omega = k\}] &= \mathbf{P}\left[\bigcup_{n \in \mathbb{Z}} 10^n \cdot [k, k+1)\right] \\ &= \mathbf{P}\left[\bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, k+1)\right] - \mathbf{P}\left[\bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, k)\right] \\ &= \log_{10}(k+1) - \log_{10} k = \log_{10}\left(1 + \frac{1}{k}\right) \end{aligned}$$

(d) We show that $\mathcal{M}'$ is a σ -algebra. Indeed, (S1) is clearly fulfilled since $(0, \infty) \in \mathcal{M}'$.

For (S2), let $M \in \mathcal{M}'$. Then $\alpha M \in \mathcal{M}$. Moreover, $\alpha M^c \in \mathcal{M}$, since $M^c = (0, \infty) \setminus M$ and thus $\alpha M^c = (0, \infty) \setminus \alpha M \in \mathcal{M}$, since $\mathcal{M}$ is a σ -algebra.

For (S3), consider sets $M_1, M_2, \dots \in \mathcal{M}'$. Then $\alpha \bigcup_{i=1}^{\infty} M_i = \bigcup_{i=1}^{\infty} \alpha M_i \in \mathcal{M}$, since $\mathcal{M}$ is a σ -algebra. Since $\mathcal{M}$ is the smallest σ -algebra, which contains all sets M_i , the claim follows.

2. More on σ -algebras

- (a) Give an example showing that the union $\mathcal{F} \cup \mathcal{G}$ of two σ -algebras $\mathcal{F}$ and $\mathcal{G}$ on a sample space $\Omega \neq \emptyset$ is not necessarily a σ -algebra on Ω .
- (b) Recall that the Borel- σ -algebra $\mathcal{B}(\mathbb{R})$ is defined as

$$\mathcal{B}(\mathbb{R}) = \sigma(\{[a, b]; a, b \in \mathbb{R}, a < b\}).$$

Show that for every $a, b \in \mathbb{R}$, one has $[a, b] \in \mathcal{B}(\mathbb{R})$ and $(-\infty, a] \in \mathcal{B}(\mathbb{R})$.

- (c) Show that the Borel- σ -algebra $\mathcal{B}(\mathbb{R})$ is also generated by the collection of all sub-intervals of the form $(-\infty, b]$ and the collection of all open sets in $\mathbb{R}$.

Hint: Define $\mathcal{B}_1 = \sigma(\{O \subseteq \mathbb{R}; O \text{ is open}\})$ and $\mathcal{B}_2 = \sigma(\{(-\infty, b]; b \in \mathbb{R}\})$. Then show that $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}_2 \subseteq \mathcal{B}_1$, and finally $\mathcal{B}_1 \subseteq \mathcal{B}(\mathbb{R})$. You can use the fact that every open subset O of $\mathbb{R}$ can be written as a countable union of disjoint open intervals in the last step.

- (a) Consider $\Omega = \{1, 2, 3, 4\}$ and consider the two σ -algebras

$$\begin{aligned} \mathcal{F}_1 &= \{\emptyset, \{1\}, \{2, 3, 4\}, \Omega\}, \\ \mathcal{F}_2 &= \{\emptyset, \{2\}, \{1, 3, 4\}, \Omega\}. \end{aligned}$$

The union of the two σ -algebras is

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 3, 4\}, \{2, 3, 4\}, \Omega\}.$$

This is clearly not a σ -algebra, since (S3) is violated: $\{1\}, \{2\}, \emptyset \in \mathcal{F}_1 \cup \mathcal{F}_2$, but

$$\{1\} \cup \{2\} \cup \emptyset \cup \emptyset \cup \dots = \{1, 2\} \notin \mathcal{F}_1 \cup \mathcal{F}_2.$$

- (b) Note that by definition, for every $n \in \mathbb{N}$, $[a, b + \frac{1}{n}] \in \mathcal{B}(\mathbb{R})$. Therefore,

$$[a, b] = \bigcap_{n=1}^{\infty} [a, b + \frac{1}{n}] \in \mathcal{B}(\mathbb{R}),$$

by Proposition 1.11 (ii), since $\mathcal{B}(\mathbb{R})$ is a σ -algebra. For the other set, simply note

$$(-\infty, a] = \bigcup_{n=1}^{\infty} (a - n, a] \in \mathcal{B}(\mathbb{R}),$$

by (S3) since $\mathcal{B}(\mathbb{R})$ is a σ -algebra.

- (c) We define $\mathcal{B}_1$ and $\mathcal{B}_2$ as in the hint. To show $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}_2$, by the definition of generated σ -algebra $\sigma(\cdot)$, it suffices to show that for all $a, b \in \mathbb{R}$ with $a < b$, $[a, b] \in \mathcal{B}_2$ (we will use this sufficiency implicitly in the following). Note that by definition, for every $n \in \mathbb{N}$, $(-\infty, b + \frac{1}{n}] \in \mathcal{B}_2$. Therefore, similar to (b),

$$(-\infty, b) = \bigcap_{n=1}^{\infty} (-\infty, b + \frac{1}{n}] \in \mathcal{B}_2.$$

Similarly, $(-\infty, a) \in \mathcal{B}_2$. Hence $[a, b] = (-\infty, b) \setminus (-\infty, a) \in \mathcal{B}_2$, and thus $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}_2$. We then proceed to show $\mathcal{B}_2 \subseteq \mathcal{B}_1$. As $(-\infty, b] = (b, \infty)^c \in \mathcal{B}_1$, the inclusion is clear. It remains to show $\mathcal{B}_1 \subseteq \mathcal{B}(\mathbb{R})$. Note that by definition, for every $a, b \in \mathbb{R}$ with $a < b$ and $n \in \mathbb{N}$, $n \in [a + \frac{1}{n}, b) \in \mathcal{B}(\mathbb{R})$. So

$$(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b) \in \mathcal{B}(\mathbb{R}).$$

Similarly, $(-\infty, a) \in \mathcal{B}(\mathbb{R})$ and $(b, \infty) \in \mathcal{B}(\mathbb{R})$. As every open subset of $\mathbb{R}$ can be written as a countable union of disjoint open intervals, we conclude $O \in \mathcal{B}(\mathbb{R})$.

3. Calculations with continuous distributions

- (a) Suppose the life-time of a lightbulb is exponentially distributed with parameter $\lambda > 0$, where time is measured in years.
- What is the probability that the lightbulb lasts for at least 2, but at most 3 years? Give a formula for general λ and then specify it to $\lambda = 1_{\text{Year}}$.
 - For general $\lambda > 0$, calculate the half-life, that is the value $t > 0$ such that the probability for the lightbulb to last at least until time t is exactly $\frac{1}{2}$.

- (b) Consider again the exponential distribution $\mathbf{P} = \mathcal{E}(\lambda)$. Show that it satisfies the memoryless property, i.e. show that for $s, t > 0$, one has

$$\mathbf{P}[(s+t, \infty) | (s, \infty)] = \mathbf{P}[(t, \infty)].$$

- (c) The *continuous Laplace²-distribution with parameter $\lambda > 0$* is a probability measure $\mathbf{P}$ on $\mathbb{R}$ characterized by the density

$$f(x) = c_{\lambda} \exp\left(-\frac{|x|}{\lambda}\right), \quad x \in \mathbb{R}.$$

- Determine the constant c_{λ} .
- Determine for the cumulative distribution function $F(x)$ for $x \in \mathbb{R}$.
- Calculate $\mathbf{P}([-1, 1])$.

- (a) i. The probability we need to calculate is

$$\mathbf{P}[[2, 3]] = \int_2^3 \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_2^3 = e^{-2\lambda} - e^{-3\lambda} = \frac{1}{e^{2\lambda}} \left(1 - \frac{1}{e}\right).$$

If $\lambda = 1$, we have

$$\mathbf{P}[[2, 3]] = \frac{1}{e^2} \left(1 - \frac{1}{e}\right) \approx 0.0855 = 8.55\%.$$

- ii. We require

$$\frac{1}{2} = \mathbf{P}[[t, \infty)] = \int_t^{\infty} \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_t^{\infty} = e^{-\lambda t},$$

from here, it follows that

$$\lambda = \frac{\ln(2)}{\lambda}.$$

²This should not be confused with “Laplace probability space”.

(b) Note that $(s + t, \infty) \subseteq (s, \infty)$, so $(s + t, \infty) \cap (s, \infty) = (s + t, \infty)$. Now

$$\begin{aligned}\mathbf{P}[(s + t, \infty) | (s, \infty)] &= \frac{\int_{s+t}^{\infty} \lambda e^{-\lambda x} dx}{\int_s^{\infty} \lambda e^{-\lambda x} dx} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} \\ &= \int_t^{\infty} \lambda e^{-\lambda x} dx = \mathbf{P}[(t, \infty)].\end{aligned}$$

(c) i. We determine c_λ by

$$\begin{aligned}1 &= \int_{-\infty}^{\infty} f(x) dx = c_\lambda \cdot \int_{-\infty}^{\infty} \exp\left(-\frac{|x|}{\lambda}\right) dx = 2c_\lambda \cdot \int_0^{\infty} \exp\left(-\frac{x}{\lambda}\right) dx \\ &= 2c_\lambda \cdot \left[-\lambda \cdot \exp\left(-\frac{x}{\lambda}\right)\right]_0^{\infty} = 2c_\lambda \cdot \lambda.\end{aligned}$$

Then, $c_\lambda = \frac{1}{2\lambda}$. Obviously, we also have $f \geq 0$.

ii. We consider the cases $x \leq 0$ and $x > 0$ separately.

• For $x \leq 0$, we have

$$\begin{aligned}F(x) &= \int_{-\infty}^x f(y) dy = c_\lambda \int_{-\infty}^x \exp\left(\frac{y}{\lambda}\right) dy = c_\lambda \lambda \left[\exp\left(\frac{y}{\lambda}\right)\right]_{-\infty}^x \\ &= c_\lambda \cdot \lambda \exp\left(\frac{x}{\lambda}\right) = \frac{1}{2} \exp\left(\frac{x}{\lambda}\right).\end{aligned}$$

• For $x > 0$ (recall that $F(0) = \frac{1}{2}$), we have

$$\begin{aligned}F(x) &= \int_{-\infty}^x f(y) dy = \frac{1}{2} + c_\lambda \int_0^x \exp\left(-\frac{y}{\lambda}\right) dy = \frac{1}{2} + c_\lambda \left[-\lambda \exp\left(-\frac{y}{\lambda}\right)\right]_0^x \\ &= 1 - \frac{1}{2} \exp\left(-\frac{x}{\lambda}\right).\end{aligned}$$

iii. We use the cumulative distribution function:

$$\begin{aligned}\mathbf{P}[-1, 1] &= F(1) - F(-1) = \left(1 - \frac{1}{2} \exp\left(-\frac{1}{\lambda}\right)\right) - \left(\frac{1}{2} \exp\left(-\frac{1}{\lambda}\right)\right) \\ &= 1 - \exp\left(-\frac{1}{\lambda}\right).\end{aligned}$$

4. (*) On probability density functions

[4 Points]

(a) Let

$$f(x) := \frac{1}{x^k} \mathbb{1}_{[1, +\infty)}(x).$$

For what value of $k \in \mathbb{R}$, if any, is f a probability density function?

(b) Give an example of a probability density function f such that $c\sqrt{f}$ cannot be a probability density function for any $c > 0$.

(c) Let

$$f(x) = 2|x| (1 - x^2) \mathbb{1}_{\{|x| \leq 1\}}.$$

i. Find the corresponding cumulative distribution function F .

ii. Let $\mathbf{P}$ be the associated probability measure. Compute $\mathbf{P}((-\infty, -\frac{1}{2}))$ and $\mathbf{P}([-\frac{1}{2}, \frac{1}{2}])$.

(a) We know that for a piecewise continuous function $f \geq 0$ to be a density on $\mathbb{R}$, it has to satisfy

$$\int_{-\infty}^{\infty} f(t) dt = 1.$$

We see that for f to be at all integrable, k has to be strictly larger than 1. Let $k > 1$. Then

$$\int_{-\infty}^{\infty} \frac{1}{x^k} \mathbb{1}_{[1, \infty)}(x) dx = \int_1^{\infty} \frac{dx}{x^k} = \frac{1}{k-1}.$$

Then $k-1 = 1 \Rightarrow k = 2$. Hence, $k = 2$ is the only possibility for f to be a density.

(b) If we take

$$f(x) = \frac{1}{x^2} \mathbb{1}_{x \in [1, +\infty)},$$

then

$$\sqrt{f(x)} = \frac{1}{x} \mathbb{1}_{x \in [1, +\infty)}$$

which is not integrable, which implies that there is no $c \in \mathbb{R}$ such that $c\sqrt{f}$ is a density.

(c) i. By definition, the cumulative distribution function corresponding to f is given by

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt \\ &= \begin{cases} 0 & \text{if } x < -1, \\ 2 \int_{-1}^x |t|(1-t^2) dt & \text{if } -1 \leq x < 1, \\ 1 & \text{if } x \geq 1. \end{cases} \end{aligned}$$

In the interval $[-1, 1]$, there are two further cases:

$$F(x) = \begin{cases} 2 \int_{-1}^x (-t)(1-t^2) dt & \text{if } -1 \leq x \leq 0, \\ 2 \int_{-1}^0 (-t)(1-t^2) dt + 2 \int_0^x t(1-t^2) dt & \text{if } 0 \leq x \leq 1, \end{cases}$$

where

$$2 \int_{-1}^x (-t)(1-t^2) dt = \left[\frac{(1-t^2)^2}{2} \right]_{-1}^x = \frac{(1-x^2)^2}{2}$$

and

$$2 \int_0^x t(1-t^2) dt = \left[-\frac{(1-t^2)^2}{2} \right]_0^x = \frac{1}{2}(1 - (1-x^2)^2).$$

Thus

$$F(x) = \begin{cases} \frac{(1-x^2)^2}{2} & \text{if } -1 \leq x \leq 0, \\ 1 - \frac{(1-x^2)^2}{2} & \text{if } 0 \leq x \leq 1. \end{cases}$$

To conclude,

$$F(x) = \begin{cases} 0 & \text{if } x < -1, \\ \frac{(1-x^2)^2}{2} & \text{if } -1 \leq x < 0, \\ 1 - \frac{(1-x^2)^2}{2} & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x \geq 1. \end{cases}$$

ii. We have

$$\begin{aligned} \mathbf{P} \left[\left(-\infty, -\frac{1}{2} \right) \right] &= \mathbf{P} \left[\left(-\infty, -\frac{1}{2} \right] \right] \\ &= F \left(-\frac{1}{2} \right) \\ &= \left(1 - \frac{1}{4} \right)^2 \times \frac{1}{2} \\ &= \frac{9}{32} \end{aligned}$$

(in the first step we used the fact that $\mathbf{P}$ is a continuous distribution, implying that $\mathbf{P}(\{a\}) = 0$ for any $a \in \mathbb{R}$). For the second probability,

$$\begin{aligned}
 \mathbf{P}\left[-\frac{1}{2}, \frac{1}{2}\right] &= \mathbf{P}\left[-\frac{1}{2}, \frac{1}{2}\right] \\
 &= F\left(\frac{1}{2}\right) - F\left(-\frac{1}{2}\right) \\
 &= 1 - F\left(-\frac{1}{2}\right) - F\left(-\frac{1}{2}\right) \\
 &= 1 - 2 \times \frac{9}{32} \\
 &= 1 - \frac{9}{16} = \frac{7}{16}.
 \end{aligned}$$